

Mohammadreza Nemati

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EXPERIENCE

AI Engineer

05/2025 - Present

Volkswagen Group of America

Belmont, CA

- Implemented depth-pruning for Qwen2.5-VL by constructing 8L, 6L, and 4L student variants, mapping selected teacher layers and preserving compatible vision-language weights for constrained deployment.
- Distilled knowledge from Qwen2.5-VL 7B into the pruned 1.4B trajectory-prediction model, applying quantization-aware training to preserve VLM performance under constrained deployment settings.
- Built multi-task VLM fine-tuning pipelines for automotive HMI reasoning, combining toggle-state prediction, step decomposition, and UI grounding with DDP training and task-specific evaluation.
- Post-trained multimodal foundation models using SFT, DPO, and GRPO for reasoning, grounding, and task execution in automotive AI workflows.
- Deployed and evaluated Qwen3.5 models up to 35B/A3B on Azure ML managed endpoints with vLLM, configuring H100/A100 serving, prefix caching, long-context inference, health probes, and throughput/latency monitoring.

Artificial Intelligence Research Assistant

08/2022 - 06/2025

Case Western Reserve University

Cleveland, Ohio

- Improved LLM KV-cache quantization by analyzing key/value norm imbalance, achieving $2\times$ memory reduction with $>94\%$ accuracy retention across models up to 70B parameters, directly optimizing inference memory efficiency for large-scale serving [\[paper link\]](#)
- Developed LIT-LVM using latent variable embeddings, structured regularization, and GNN-based higher-order interactions, outperforming GNN, Factorization Machine, and elastic net baselines in ranking and recommendation tasks [\[paper link\]](#)
- Developed DEASurv, a deep multimodal model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve prediction on large-scale medical datasets

Software Engineer - AI/ML Intern

02/2024 - 07/2024

Sherwin-Williams

Cleveland, Ohio

- Designed and deployed real-time inference pipeline for vision transformers (SAM), optimizing model latency and improving segmentation quality by 7% IoU through inference and evaluation improvements.
- Built multimodal system integrating LLaVA with vision transformers, implementing cross-modal alignment for natural-language visual grounding and object segmentation.

Data Science Research Intern

05/2023 - 09/2023

Beyond Limits AI

Glendale, California

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 using PyTorch distributed training with FSDP/DDP for multi-GPU optimization and quantization for efficient deployment.

SKILLS

Programming: Python, C++

Foundation Models: Multimodal Foundation Models, LLMs/VLMs, Transformers, Vision Transformers, Fine-tuning & Post-training, Reinforcement Learning

Optimization: Knowledge Distillation, Quantization, Pruning, CUDA, Continuous Batching

ML Systems: PyTorch, vLLM, Distributed Training (FSDP/DDP), Model Serving, Experiment Design, Model Evaluation

EDUCATION

Case Western Reserve University – PhD, Computer Science

Cleveland, OH | 2025

University of Toledo – MS, Computer Science

Toledo, OH | 2020

Shiraz University – BS, Electrical Engineering

Shiraz, Iran | 2017